

# RIYA MEHTA

 riyagmehta

 riya-mehta

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 Chicago, IL

## EDUCATION

### University of Illinois Chicago

Master of Science in Computer Science

August 2024 – May 2026

GPA: 4.0/4.0

### Dharmsinh Desai University

Bachelor of Technology in Information Technology

Aug 2020 – May 2024

CGPA: 8.7/10.0

## SKILLS

**Languages:** Java, Python, TypeScript, JavaScript, SQL

**AI & LLM:** LLMs, AI Agents, RAG, Embeddings, MCP, Structured Outputs, Model Evaluation

**Backend:** FastAPI, Node.js, Express, Spring Boot, Rust/Actix, REST APIs, Microservices

**Frontend:** React, Next.js, Redux, Zustand, Jotai, Tailwind CSS, Chakra UI

**Data:** PostgreSQL, MySQL, MongoDB, SQLite, Redis, Pinecone, MeiliSearch, IndexedDB

**Cloud & Engineering:** AWS, Docker, nginx, Oracle Cloud Infrastructure, GitHub Actions, CI/CD, Playwright, Postman

## EXPERIENCE

### Pramata

AI Solution Engineering Intern

July 2026 – Present

Remote, USA

- Design and ship GenAI solutions for enterprise contract workflows, combining retrieval, tool execution, validation, and structured outputs across multiple LLM providers.
- Build reusable agent components and MCP plugins that expose enterprise data and application services through defined schemas, tool interfaces, and execution rules.
- Debug live customer workflows by tracing failures across retrieval, context, prompting, tool execution, and model outputs to improve reliability.

### Fuzzy Cloud

Software Developer Intern

Dec 2023 – Aug 2024

Ahmedabad, India

- Owned search for a production healthcare application, integrating MeiliSearch with debounced fuzzy queries to reduce response time by 35% for incomplete and misspelled input.
- Built offline-first record access with IndexedDB, synchronizing local writes through CQRS and event-sourcing workflows without blocking the UI on network state.
- Developed client state management with Zustand and Jotai across a Rust/Actix backend and SQLite persistence layer.
- Reproduced and resolved production failures across frontend, API, search, and persistence using Playwright and Postman.

### Oasis Infobyte

Web Development Intern

July 2023 – Aug 2023

Remote, India

- Designed REST APIs for order creation, status updates, and user management and integrated them with React workflows backed by persistent application state.
- Built multi-step order processing with validation and state transitions across React, Express, and database layers, debugging issues across the full request path.

## PROJECTS

### Distributed API Gateway

2026

Node.js, TypeScript, PostgreSQL, Docker, nginx, Oracle Cloud Infrastructure

- Built a distributed API gateway with an atomic PostgreSQL token-bucket rate limiter, verified under concurrent requests across multiple gateway instances with negative-control tests for race conditions.
- Implemented closed, open, and half-open circuit-breaker states to isolate failing downstream services and recover automatically after health restoration.
- Added Prometheus metrics for request rates, latency percentiles, and breaker state; deployed multiple gateway instances behind nginx with shared PostgreSQL state on OCI.

### AI-Powered Adaptive Course Generation Platform

Dec 2025

FastAPI, React, TypeScript, PostgreSQL, Redis, ARQ, OpenAI, Pinecone

- Built durable asynchronous course generation with Redis-backed ARQ jobs, retries, exponential backoff, idempotency, and crash recovery for unfinished module generation.
- Built RAG-based Q&A with OpenAI embeddings, Pinecone retrieval, and Redis caching; evaluated recall@1/3/5, MRR, groundedness, and refusal rate across 25 test cases.
- Instrumented API, background workers, and LLM calls with request tracing, token usage, cost, and latency metrics to diagnose performance and reliability issues.